Answer Key — The Magnetic Effect of Electric Current

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

Q1 **(b) deflects** — the current's magnetic field deflects the needle.

Q2 **(c) Oersted** — Hans Christian Oersted, in 1820.

Q3 **(b) magnetic field** — that region is the magnetic field.

Q4 **(b) False** — the field disappears when the current stops.

Q5 **(a) Both true, R explains A** — the field produced by the current is exactly why the needle deflects.

Q6 **magnetic** — this is the magnetic effect of electric current.

Q7 **1820** — Oersted's discovery was in 1820.

Short answer

Q8 ON: the compass needle deflects (current makes a magnetic field). OFF: the needle returns to its original direction (field gone).

Q9 The region around a magnet or current-carrying wire where its magnetic effect can be felt. Device: electromagnet / electric bell / motor / fan / loudspeaker (any one).

Q10 That electricity and magnetism are linked — he saw a compass needle deflect whenever a nearby circuit was switched on/off.

Long / application

Q11 Place a compass under a straight wire connected to a cell through a switch. Switch ON: the needle deflects; switch OFF: it returns. **Conclusion:** a current produces a magnetic field, i.e. current has a magnetic effect.

Q12 Reversing the terminals reverses the current direction, which reverses the direction of the magnetic field, so the needle deflects to the opposite side.

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